

Private Payments

Omid Yaghoobi, Koosha Kazemi

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Pros & Cons

01 - Motivation

- Ideally, any two transacting parties, Alice and Bob should remain:
 - Unlinkable and Unobservable
 - to the rest of the network
 - and optionally even to each other – unless they explicitly choose not to

02 - Unsatisfying Solutions

- Ignoring it
 - share a static address
- Manual sharing
 - generate and communicate new address every time
- Extended public keys (xpub)
 - Each xpub can be shared with only a single partner
- Addresses on-demand (Using a payment server)
 - Services like BTCPay take care of the communication
 - Mostly for business users + a centralized server is vulnerable
- So...?
 - Use a communication channel which is going to be used anyway: the Bitcoin network

03 - BIP-47: Payment Codes

- Recipient has a public, static, reusable payment code
 - A pubkey, the notification address
- Sender sends a **notification transaction** to introduce themselves
 - Perform ECDH to calculate a shared secret
 - Use the secret to encrypt and embed their own payment code
 - Only needs to happen once
- ECDH used again to generate unique payment addresses
- **Pros:**
 - Preserves privacy
 - Non-interactive after setup
 - Contacts

03 - BIP-47: Payment Codes (cont.)

- **Cons:**
 - Notification address is reused → visible linkability and observability
 - Will be revisited shortly
 - does not say anything about address types → have to watch all address types, might not be able to spend
 - Additional on-chain TX → extra fee paid for privacy



PayNyms

04 - Private Payments (BIP-351)

- Builds on the core ideas of BIP-47
- Address type also embedded in payment code and notification transaction
- Removes the need for a reused notification address
 - Why is this such a big deal? Let's see an example
 - WikiLeaks doesn't want governments to observe its activity
 - BIP-47 notifications are out in the wild, Anyone can observe
 - Some politician might want to donate to WikiLeaks but stay anonymous
 - If he doesn't engage in coin control, identity is exposed forever
 - BIP-351 solves this by encrypting the locking script

04 - Private Payments (BIP-351) (cont.)

- How exactly?:
 - Alice encrypts notification data using a shared secret
 - Data is embedded in an **OP_RETURN** output
 - Only Bob can decrypt this and see Alice's pubkey
 - No one knows that only Bob can do this
- Caveat: Bob has to scan all TXs on the network that start with **OP_RETURN**
 - Requires access to full blocks → computationally intensive

05 - BIP-351: Notifications (Alice)

1. Assigns an unused index x to Bob
2. Calculates a 4-Byte notification code = $H(n_x * P)[0...4]$
3. Chooses an address type supported by Bob's payment code
4. Constructs the notification payload
 - $\text{payload} = \text{"PP"} \mid \text{notification_code} \mid N_x \mid \text{address_type}$
5. Selects any UTxO in her wallet, preferably not associated with her
6. Sends a transaction including an **OP_RETURN** with a value of payload

- P : public key contained in Bob's payment code
- p : private key associated with Bob's public key P
- N : extended public key used by Alice to derive child keys for each Bob she wants to transact with
- n : private key associated with Alice's public key N
- x : Alice's secret recipient index, unique for each Bob
- N_x : child public key derived from N at index x (non-hardened)
- n_x : private key associated with N_x

05 - BIP-351: Notifications (Bob)

1. Breaks down the payload into its four constituent parts
2. Discards the `search_key` = "PP"
3. Extracts `N_x` and performs $H(N_x * p)$
 - ECDH implies that $N_x * p = n_x * P$
4. If the first four bytes match the `notification_code` then notification is valid
5. Otherwise, Bob discards notification as spurious

- P: public key contained in Bob's payment code
- p: private key associated with Bob's public key P
- N: extended public key used by Alice to derive child keys for each Bob she wants to transact with
- n: private key associated with Alice's public key N
- x: Alice's secret recipient index, unique for each Bob
- N-x: child public key derived from N at index x (non-hardened)
- n-x: private key associated with N-x

06 - BIP-351: Transacting

Alice initializes counter c unique to Bob, incremented with each TX

1. Alice calculates a secret point:
 - $S = n_x * P$
2. Alice calculates a shared secret:
 - $s = H(S \parallel c)$
3. Alice calculates Bob's ephemeral public key and its associated address where the funds will be sent
 - $P_c = P + s * G$
4. Bob constructs his watchlist by mirroring this process on his end, except he calculates S as follows:
 - $S = N_X * p$
5. When Bob wants to spend from such addresses, his private key would be:
 - $p_c = p + s$

07 - BIP-351: Pros & Cons

- **Pros**

- No notification address reuse → improved privacy.
- Non-interactive after setup (like BIP-47).
- Scales better than BIP-352 (less scanning, only those with starting **OP_RETURNs**).
- Compatible with many existing wallet primitives.
- an external service can include Alice's notification in a transaction completely unrelated to Alice's wallet
 - Decrease fees, help preserve privacy

- **Cons**

- Requires all wallets to implement a new notification flow.
- Slightly more complex logic (ECDH, decryption, address derivation).
- Recipients must monitor all **OP_RETURNs** (though this is lighter than full output scanning).
- Not yet widely adopted → ecosystem immaturity.

08 - Related and Future Work

- Silent Payments (BIP-352)
 - A more aggressive approach — no notification, heavier scanning.
 - Will be presented by another team
- Could be combined with other proposals (e.g., adapter signatures, PTLCS).
- Wallet adoption and standards work ahead.

09 - Conclusion

- A recipient that wishes to receive funds privately has several options
- Each solution has tradeoffs in terms of chain analysis potential, recoverability, and wallet complexity
- BIP-47 addresses most issues, except for the public static notification address
- BIP-351 builds on top of BIP-47
 - Addresses the remaining issues
 - Introduces new challenges (i.e., full block necessity)
- Not yet widely adopted, but seems promising
- BIP-352, an alternative approach, will be presented shortly

Thanks!

- Omid Yaghoobi
 - oysamani@gmail.com
- Koosha Kazemi
 - kooshakazemi@gmail.com

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References

- <https://blog.bitbox.swiss/en/reusable-payment-codes-exchanging-bitcoin-addresses-privately/>
- <https://github.com/bitcoin/bips/blob/master/bip-0351.mediawiki>
- <https://github.com/bitcoin/bips/blob/master/bip-0047.mediawiki>
- <https://github.com/bitcoin/bips/blob/master/bip-0032.mediawiki>
- <https://gist.github.com/RubenSomsen/c43b79517e7cb701ebf77eec6dbb46b8#user-content-fn-3-0e1cfac55a5f6515afa73771f736e880>
- <https://silentpayments.xyz/docs/comparing-proposals/bip351/>